Hospital Management Automation

n8n Workflow Documentation

Technical documentation of the four supplied n8n workflows: Appointment Booking, OCR / Medical Report Analysis, Doctor Consultation Reminder, and Admin Management / Analytics.

Documentation basis: The actual n8n workflow JSON files supplied with the project.

1. Project Overview

This project is an n8n-based hospital-management automation prototype containing four independent workflows:

• **Appointment Booking** — collects patient details, checks whether the requested date/time already exists, requests administrator approval, stores approved bookings, and sends confirmation/rejection email.

• **OCR / Medical Report Analysis** — watches a Google Drive folder for newly created files, downloads the file, sends it to OCR.space, structures extracted text using a Gemini-powered AI Agent, and emails an HTML analysis.

• **Doctor Consultation Reminder** — records follow-up requirements and dates, then periodically checks stored follow-ups and sends reminder emails.

• **Admin Management / Analytics** — periodically reads hospital booking data, calculates basic appointment statistics, asks Gemini to generate an administrative report, and emails the report.

The implementation uses n8n Data Tables as the current data store, Gmail for email communication, Google Drive for document intake, OCR.space for OCR, and Google Gemini for AI-generated analysis/reporting.

2. Technology Stack

Layer	Technology	Role
Automation engine	n8n	Executes and connects all workflows
Workflow logic	n8n Code / JavaScript	Validation, counting and JSON parsing
Forms	n8n Form Trigger	Patient booking and consultation input
Database / storage	n8n Data Tables	Appointment and consultation records
File storage/input	Google Drive	Medical-report file intake
OCR	OCR.space REST API	Converts uploaded document/image into text
AI	Google Gemini via n8n AI Agent	Structures OCR output and creates admin reports
Email	Gmail	Approval, confirmations, reports and reminders
Scheduling	n8n Schedule Trigger	Periodic analytics and reminder processing

3. Workflow 1 — Appointment Booking

Purpose

Automates the appointment-request process from patient data collection through availability checking, administrator approval, database insertion, and email confirmation.

Trigger and Inputs

n8n Form Trigger — On form submission. The form is titled **Patient Details**.

• Patient name — required

• Age — required numeric field

• Sex — required radio selection (male / female)

• Phone number — required numeric field

• Appointment Date

• Time — Morning / Evening

Node-by-node flow

• **On form submission:** receives patient and appointment details.

• **checking appointment:** queries the n8n Data Table named *hospital mangement* and checks Date and Time.

• **Code in JavaScript:** converts the lookup result into an *exists* boolean.

• **If:** branches according to availability.

• **Send message and wait for response:** Gmail sends an approval request to the configured administrator.

• **If1:** checks *$json.data.approved*.

• **Insert row:** stores the approved appointment in the n8n Data Table.

• **Send a message1:** sends the appointment-booked confirmation email.

```
Patient
  ↓
Patient Details Form
  ↓
Check Date + Time in Data Table
  ↓
Is slot already occupied?
  ■■■ YES → Rejection Email
  ■■■ NO
       ↓
   Admin Approval Request
       ↓
   Approved?
   ■■■ NO → Rejection Email
   ■■■ YES
       ↓
   Insert Appointment
       ↓
   Confirmation Email
```

Input / Output

Input: patient and appointment details. **Output:** slot rejection, approval/rejection outcome, or stored appointment plus confirmation email.

4. Workflow 2 — OCR / Medical Report Analysis

Purpose

Automatically processes newly uploaded medical-report files and converts OCR output into a structured report using Gemini.

Trigger

Google Drive Trigger — Google Drive Trigger. It watches a specified Google Drive folder and checks periodically for newly created files.

Node-by-node flow

• **Google Drive Trigger:** detects a newly created file.

• **Download file:** downloads the file from Google Drive using its file ID.

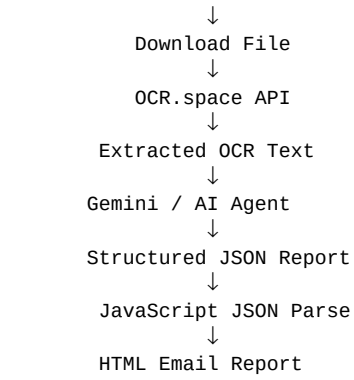
• **HTTP Request:** sends the binary file to the OCR.space Parse Image API using multipart form data.

• **AI Agent:** uses the OCR result and Gemini to return structured JSON containing patientName, tests, and summary.

• **Code in JavaScript:** removes accidental Markdown code fences and parses the AI response.

• **Send a message:** creates an HTML email containing patient name, tests, values, reference ranges, status and summary.

```
Medical Report Uploaded to Google Drive
                ↓
       Google Drive Trigger
```



AI Output Structure

```
{
  "patientName": "",
  "tests": [
    {
      "testName": "",
      "value": "",
      "unit": "",
      "referenceRange": "",
      "status": ""
    }
  ],
  "summary": ""
}
```

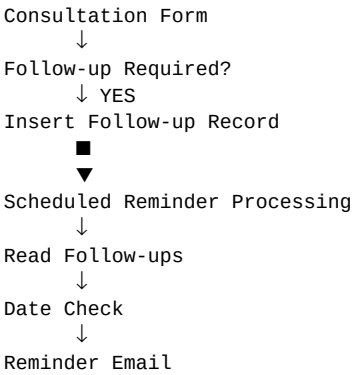
5. Workflow 3 — Doctor Consultation Reminder

Purpose

Records patients who require a follow-up consultation and periodically checks stored records for reminder processing.

Trigger A — Consultation Form

n8n Form Trigger — On form submission. Inputs include Name, Gmail, Follow-up Required (Yes/No), and Follow-up Date.



Storage

The follow-up record is stored in the n8n Data Table named *consultation*, with Name, Gmail, consult_required and Date.

Trigger B — Scheduled Reminder Processing

• **Get row(s):** reads consultation records.

• **If1:** checks whether the stored follow-up date meets the configured date condition.

• **Send a message:** sends the Doctor Consultation Reminder email.

6. Workflow 4 — Admin Management / Analytics

Purpose

Generates a periodic hospital-management analytics report from stored records.

Node-by-node flow

• **Schedule Trigger:** starts the analytics workflow periodically.

• **Get row(s):** reads records from the n8n Data Table.

• **Code in JavaScript:** calculates total patients, appointments booked and appointments not booked.

• **AI Agent:** uses Gemini to create a short management report including summary, booking percentage, patients without appointments and an administrative observation.

• **Send a message:** emails the generated Hospital Analytics Report.

```
Weekly Schedule Trigger
          ↓
    Read Data Table
          ↓
    JavaScript Statistics
          ↓
     Gemini AI Agent
          ↓
 Administrative Report
          ↓
       Gmail Email
```

7. Data Stores

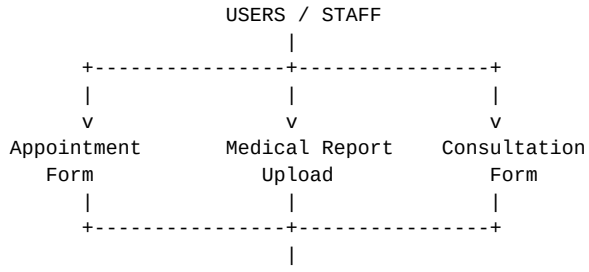
Data Table	Current role	Fields / information
hospital mangement	Appointment workflow	Name, Age, Sex, phone, Date, Time
consultation	Reminder workflow	Name, Gmail, consult_required, Date

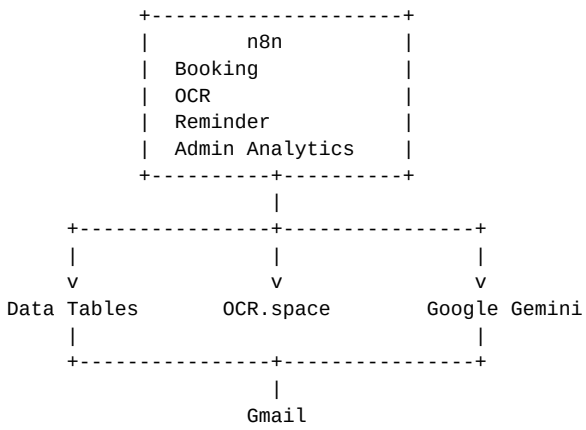
8. External Services

Service	Purpose
Google Drive	Receives medical-report files and triggers OCR processing
OCR.space	Performs OCR on uploaded document/image files
Google Gemini	Structures OCR text and generates management analytics
Gmail	Approval, confirmation, reminder and report emails

9. System Architecture

The current system is best represented as four modular n8n automations connected to the project's actual external services and n8n Data Tables.





A separate PNG architecture diagram is supplied with this documentation and can be inserted into a report or presentation.

10. Important Implementation Notes Before Submission

• **OCR API key:** The supplied OCR workflow contains an API key directly inside the HTTP Request configuration. Revoke/rotate it and move the replacement into n8n credentials or secret management before public GitHub release.

• **Hard-coded email addresses:** Several Gmail nodes contain fixed recipient addresses. Use credentials, environment variables, or dynamic recipient fields instead of publishing personal addresses.

• **Reminder recipient:** The consultation form collects a Gmail field, but the current reminder email configuration should be checked so it dynamically uses the stored patient's email.

• **Reminder date logic:** The scheduled branch should be tightened to define exactly when a reminder is sent and prevent duplicate reminders, for example using a reminder_sent field.

• **Appointment lookup edge case:** The availability workflow should explicitly handle the case where the Data Table returns zero matching rows.

• **Admin analytics schema:** The analytics JavaScript expects an appointment field with values such as yes/no, while the current booking insert mapping shown in the supplied workflow should be checked for that field.

11. Recommended Production Architecture

For a stronger production version, the prototype could evolve toward:

• n8n for orchestration

• PostgreSQL for persistent structured data

• Google Drive or object storage for documents

• OCR service for text extraction

• Gemini for structured extraction and summarization

• Gmail/SMS service for notifications

• role-based admin access

• secret management for API credentials

• audit logs

• reminder status fields to prevent duplicate notifications

These are improvement recommendations; they are not all present in the supplied workflow JSON files.

12. Overall Workflow Inventory

Workflow	Trigger	Main processing	Storage	External services	Output
Appointment Booking	Form submission	Availability + approval	n8n Data Table	Gmail	Booking/rejection email
OCR	Google Drive file created	OCR + Gemini structuring	No persistent report table in current flow	Drive, OCR.space, Gemini, Gmail	Email report
Reminder	Form + schedule	Follow-up storage + date check	n8n Data Table	Gmail	Reminder email
Admin Analytics	Weekly schedule	Counting + Gemini report	n8n Data Table	Gemini, Gmail	Analytics email

13. Conclusion

The supplied project is a four-workflow hospital automation prototype built around n8n. Its core idea is to automate repetitive hospital operations rather than build a conventional monolithic application: appointment management, medical-document OCR, follow-up reminders, and administrative analytics.

Core concept: Appointment Management + Medical-document OCR + Follow-up Reminders + Administrative Analytics, orchestrated through n8n.